Meng Zheng

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Google Scholar: https://scholar.google.com/citations?user=1D5PfMgAAAAJ&hl=en

My research focuses on the design and development of interpretable deep learning systems and algorithms for a variety of computer vision and medical imaging applications, with particular focus on automated patient positioning and 2D/3D human modeling, Vision-Language-Action (VLA) models, generative modeling and image retrieval/person re-identification.

EDUCATION

Rensselaer Polytechnic Institute — **Troy, NY**
PhD in Electrical Engineering, Adviser: Prof. Richard J. Radke, GPA: 3.94 — *May 2020*

Beijing Institute of Technology — **Beijing, China**
Master of Science in Electrical Engineering, GPA: 3.89 — *March 2016*

Beijing Institute of Technology — **Beijing, China**
Bachelor of Electrical Engineering, GPA: 3.90 — *March 2013*

WORK EXPERIENCE

UII America, Inc. — **Burlington, MA**
Expert Research Scientist — *January 2024 – present*

- **Robotic Skill Acquisition with Foundation Models**
  Developed a language-guided robot skill acquisition framework leveraging Diffusion and Flow-Matching objectives, conditioned on dynamic feedback from execution failures. Designed a dual-expert learning strategy using a shared vision-language model (VLM) backbone to jointly optimize success and failure action experts on a curated mixture of trajectories. Enabled autonomous error discovery to capture a broader distribution of failure modes, improving policy robustness and refinement. Deployed the framework on an ultrasound robotic arm prototype.
- **Robotic Ultrasound Analytics**
  Built a robust visuomotor policy framework upon high-fidelity Isaac Sim environment with a real-time ROS2/MoveIt2 bridge, enabling large-scale data collection from simulation and real robotic ultrasound systems. Developed a novel direction-controlled Diffusion Policy for precise, input-guided motion and deployed it as a low-latency, multimodal Flask service for real-time inference. The vendor-agnostic architecture generalizes seamlessly across KUKA and proprietary Kylin robotic arms.
- **Occlusion-free Monocular Clothed Human Reconstruction**
  Developed a novel 3D human modeling pipeline that generates occlusion-resilient and multiview-consistent 3D humans from a single image without requiring 3D ground-truth data. The proposed pipeline adopts two-stage (Multiview Diffusion Synthesis and 3DGS Reconstruction) pipeline which outperforms existing methods in both novel view synthesis and geometric accuracy under challenging real-world conditions. Work published as **Highlight** in *IEEE/CVF International Conference on Computer Vision* (***ICCV***) 2025.
- **Image/Video Few Shot Segmentation (FSS) for Smart Annotation Tool**
  Developed a novel Transformer-based deep network with a lightweight multi-agent dense fusion (MADF) module, to extensively identify the complementary and consistent information among multiple supports, for accurately segmenting the object of interest in the query, with +3% mean Intersection over Union (mIoU) increase on average with virtually no model parameter increase

compared to existing state-of-the-art methods on two benchmark FSS datasets. Additionally, we show how our proposed module can be extended to a new under-explored problem, the effective segmentation of an unseen object in a query video given a few support videos, by considering dense appearance as well as motion correspondences. The developed algorithm can be directly applicable for cross-domain 3D segmentation. Early accepted at *International Conference on Medical Image Computing and Computer Assisted Intervention* (***MICCAI***) 2024.

- **Cross-Modality Domain Adaptation (CMDA) framework for Nighttime Semantic Segmentation**
  Participated in developing a novel unsupervised CMDA framework to leverage multi-modality (Images and Events) information for nighttime semantic segmentation, with only labels on daytime images. In CMDA, we design the Image Motion-Extractor to extract motion information and the Image Content-Extractor to extract content information from images, in order to bridge the gap between different modalities (Images $\rightleftharpoons$ Events) and domains (Day $\rightleftharpoons$ Night). Work accepted by *IEEE/CVF International Conference on Computer Vision* (***ICCV***) 2023.

UII America, Inc. **Cambridge, MA**
Senior Research Scientist *June 2020 – December 2023*

- **Contactless Patient Modeling and Positioning System**
  Developed fully automated, contactless patient modeling and positioning systems for different types of medical scanners and smart scanning/operation rooms, e.g. Computed Tomography (CT), Molecular imaging (MI), Magnetic Resonance Imaging (MRI) or Digital subtraction angiography (DSA). The system assists scanners to automatically localize patient and adjust scanner pose for more efficient and accurate medical scanning (comparing to manual operation and competitive solutions), reduced patients' radiation intake, and minimized technician's risks w.r.t infectious diseases.
- **Self-Supervised 3D Patient Modeling with Multi-modal Attentive Fusion**
  Developed a novel 3D patient modeling and positioning deep learning-based algorithm, with a robust multi-modal (RGBD) patient body joint localization module and a self-supervised 3D mesh reconstruction module, which outputs a parameterized mesh representation of the patient body. The algorithm is robust to challenging real-clinical scanning/operating scenarios, e.g. the patient is heavily covered by surgical sheet or lead vest, that most of the existing solutions would be significantly affected due to collapsed visual cues. The developed 3D mesh reconstruction algorithm does not require 3D mesh annotations for deep model training, which are essential to existing solutions and very expensive to collect, thus significantly reduces human labor effort and costs in real-clinical deployment.
- **Smart Annotation Tool**
  Developed a smart medical data annotation system with a novel weakly-supervised few-shot segmentation algorithm module, that is able to automatically propagate annotation from user-annotated data to unlabelled data without requiring intra-domain annotated training data, which sufficiently reduces user labeling time and effort comparing to conventional data annotation tools. The designed annotation algorithm is generic and applicable to different domains of medical data, e.g., brain MRI, lung CT data, etc.
- **Head Motion Detection and Correction for PET/CT Imaging**
  Developed vision-based head motion detection algorithm to automatically detect rigid movements of patient head and magnitude calculation, to diminish and compensate motion artifacts for Positron Emission Tomography - Computed Tomography (PET/CT) imaging and diagnosis.

UII America, Inc. **Cambridge, MA**
Research Intern *May – August 2019*

- Proposed the first gradient-based visual explanation technique for similarity CNN models, which can be applicable to a variety of vision-related applications including improved image retrieval,

weakly-supervised few shot segmentation and automated patient pose guidance. The proposed method is generic and extensible in principle to any feature embedding CNN model.
- Designed the first deep architecture able to visually explain the model prediction of a certain patient pose (for positioning applications) as “correct" or “incorrect" to the reference poses.
- Designed a novel workflow to provide guidance to the patient to adjust their pose according to the selected scanning protocols for better medical imaging quality based on the generated visual explanations.

Rensselaer Polytechnic Institute **Troy, NY**
Graduate Research & Teaching Assistant *Aug 2016 – May 2020*

- Substantial experience in building practical real-world systems for the problem of person re-identification (re-id)
- Designed new re-id algorithms to recover person trajectories in a multi-camera network, learning temporal models corresponding to inter-camera transition patterns, thereby improving precision in conjunction with novel search space pruning strategy to improve re-identification accuracy.
- Designed a novel video-based re-id CNN model with 3D gradient-based attention; this was the first 3D CNN-based architecture able to generate spatiotemporal localization given person class label information.
- Designed a novel two-branch architecture for image-based re-id that produces attentive regions during training without requiring any additional supervision other than identity labels or any specially-designed architecture for modeling attention.
- My work also involved collecting and annotating new datasets for the use of the research community; collected a new multi-shot multi-camera large scale re-id dataset, called RPIfield, which provides explicit time-stamp information for each candidate. The RPIfield dataset allows the study of re-id algorithms in a more general context, especially with respect to temporal aspects.
- For practical applicability, I extensively studied temporal aspects of the person re-id problem; Developed a new evaluation methodology for re-id called the Rank Persistence Curve (RPC), which provides temporal rank information derived by a re-id algorithm, showing how long a correct match to the probe is likely to stay in the rank-k shortlist
- Also have experience in the design and development of generative models; extensively studied various learning and inference algorithms for deep regression Bayesian networks under both supervised and unsupervised settings for image inpainting, image denoising, and head pose and eye gaze estimation.

SELECTED PUBLICATIONS

Journal papers

- J. Wu, Y. Xia, X. Wang, Y. Wei, A. Liu, A. Innanje, **M. Zheng**, L. Chen, J. Shi, L. Wang, Y. Zhan, X. S. Zhou, Z. Xue, F. Shi, D. Shen, “uRP: An integrated research platform for one-stop analysis of medical images", *Frontiers in Radiology*, 2023.
- X. Gong, L. Song, R. S. Vedula, A. Sharma, **M. Zheng**, B. Planche, A. Innanje, T. Chen, J. Yuan, D. Doermann, Z. Wu.“Federated Learning with Privacy-Preserving Ensemble Attention Distillation", *IEEE Transactions on Medical Imaging*, 2023.
- **M. Zheng**, S. Karanam, R. Radke.“Towards Automated Spatio-Temporal Trajectory Recovery in Wide-Area Camera Networks", *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 2020.
- S. Nie, **M. Zheng**, Q. Ji, “The Deep Regression Bayesian Network and Its Applications: Probabilistic Deep Learning for Computer Vision", *IEEE Signal Processing Magazine, Vol. 35, Issue: 1, Pages: 101-111*, 2018.
- **M. Zheng**, L. Lin. Design of Digital Temperature and Humidity Instrument Based on AT89C51, *Industry and Mine Automation, No. 6, Pages: 103-105*, 2012.

Conference papers

- **M. Zheng**, S. Marri, A. Choudhuri, B. Planche, Z. Gao, V. N. Nguyen, T. Chen, G. Chowdhary, Z. Wu, "Failing Forward: Adaptive Failure-Informed Learning for Vision-Language-Action Models", *Advances in Neural Information Processing Systems* (***NeurIPS***), 2026.
- Z. Gao, B. Planche, V. N. Nguyen, **M. Zheng**, A. Choudhuri, A. Innanje, T. Chen, Z. Wu, "Direct Conditional Parameterization for N-Dimensional Splatting", *Advances in Neural Information Processing Systems* (***NeurIPS***), 2026.
- Z. Gao, B. Planche, **M. Zheng**, A. Choudhuri, V. N. Nguyen, T. Chen, Z. Wu, "Render-FM: Feedforward Model for Real-time Photorealistic Volumetric Rendering", *European Conference on Computer Vision* (***ECCV***), 2026.
- J. Wu, V. N. Nguyen, B. Planche, J. Tao, C. Sun, Z. Gao, Z. Zhao, A. Choudhuri, G. Zhang, **M. Zheng**, F. Wang, T. Chen, Y. Yan, Z. Wu, "Consistent Instance Field for Dynamic Scene Understanding", *IEEE Conference on Computer Vision and Pattern Recognition* (***CVPR***), 2026.
- Y. Su, A. Choudhuri, Z. Gao, B. Planche, V. N. Nguyen, **M. Zheng**, Y. Shen, A. Innanje, T. Chen, E. Elhamifar, Z. Wu, "MedGRPO: Multi-task reinforcement learning for heterogeneous medical video understanding", *IEEE Conference on Computer Vision and Pattern Recognition* (***CVPR***), 2026.
- R. Liu, Z. Gao, B. Planche, M. Chen, V. N. Nguyen, **M. Zheng**, A. Choudhuri, T. Chen, Y. Wang, A. Feng, Z. Wu, "Universal Beta Splatting", *International Conference on Learning Representations* (***ICLR***), 2026.
- A. Dutta, **M. Zheng**, Z. Gao, B. Planche, A. Choudhuri, T. Chen, A. K Roy-Chowdhury, Z. Wu, "Chrome: Clothed human reconstruction with occlusion-resilience and multiview-consistency from a single image", *IEEE/CVF International Conference on Computer Vision* (***ICCV***), 2025. **Highlight** (**top 9.7%** in accepted papers, **top 2.3%** in all submitted papers)
- Z. Gao, B. Planche, **M. Zheng**, A. Choudhuri, T. Chen, Z. Wu, "7DGS: Unified Spatial-Temporal-Angular Gaussian Splatting", *IEEE/CVF International Conference on Computer Vision* (***ICCV***), 2025.
- A. Choudhuri, Z. Gao, **M. Zheng**, B. Planche, T. Chen, Z. Wu, "PolypSegTrack: Unified Foundation Model for Colonoscopy Video Analysis", *International Conference on Medical Image Computing and Computer Assisted Intervention* (***MICCAI***), 2025.
- B. Wang, A. Choudhuri, **M. Zheng**, Z. Gao, B. Planche, A. Deng, Q. Liu, T. Chen, U. Bagci, Z. Wu, "Order-aware interactive segmentation", *International Conference on Learning Representations* (***ICLR***), 2025.
- Q. Peng, B. Planche, Z. Gao, **M. Zheng**, A. Choudhuri, T. Chen, C. Chen, Z. Wu, "3D vision-language gaussian splatting", *International Conference on Learning Representations* (***ICLR***), 2025.
- Z. Gao, B. Planche, **M. Zheng**, A. Choudhuri, T. Chen, Z. Wu, "6DGS: Enhanced direction-aware gaussian splatting for volumetric rendering", *International Conference on Learning Representations* (***ICLR***), 2025.
- Z. Gao, A. Sharma, **M. Zheng**, B. Planche, T. Chen, Z. Wu, "Automated Patient Positioning with Learned 3D Hand Gestures", *IEEE/CVF Winter Conference on Applications of Computer Vision* (***WACV***), 2025.
- A. Deng, Z. Gao, A. Choudhuri, B. Planche, **M. Zheng**, B. Wang, T. Chen, C. Chen, Z. Wu, "Seq2time: Sequential knowledge transfer for video llm temporal grounding", *IEEE Conference on Computer Vision and Pattern Recognition* (***CVPR***), 2025.
- **M. Zheng**, B. Planche, Z. Gao, T. Chen, R. J. Radke, Z. Wu, "Few-Shot 3D Volumetric Segmentation with Multi-Surrogate Fusion", *International Conference on Medical Image Computing and Computer Assisted Intervention* (***MICCAI***), 2024. **Early accept** (top 11% of submissions received a provisional accept recommendation).
- Z. Gao, B. Planche, **M. Zheng**, X. Chen, T. Chen, Z. Wu, "DDGS-CT: Direction-disentangled gaussian splatting for realistic volume rendering", *Advances in Neural Information Processing Systems* (***NeurIPS***), 2024.
- W. Ren, R. Xia, **M. Zheng**, Z. Wu, Y. Tang, N. Sebe, "Cross-Class Domain Adaptive Semantic

Segmentation with Visual Language Models", *ACM Multimedia* (***MM***), 2024.

- T. Luan, Z. Gao, L. Xie, A. Sharma, H. Ding, B. Planche, **M. Zheng**, A. Lou, T. Chen, Z. Wu, "Divide and Fuse: Body Part Mesh Recovery from Partially Visible Human Images", *European Conference on Computer Vision* (***ECCV***), 2024.
- Y. Liu, B. Planche, **M. Zheng**, Z. Gao, P. Sibut-Bourde, F. Yang, T. Chen, Z. Wu, "Implicit Modeling of Non-rigid Objects with Cross-Category Signals", *Association for the Advancement of Artificial Intelligence* (***AAAI***), 2024.
- Z. Cai, Z. Gao, B. Planche, **M. Zheng**, T. Chen, M. S. Asif, Z. Wu, "Disguise without Disruption: Utility-Preserving Face De-identification", *Association for the Advancement of Artificial Intelligence* (***AAAI***), 2024.
- Z. Gao, H. Zhou, A. Sharma, **M. Zheng**, B. Planche, T. Chen, Z. Wu, "PBADet: A One-Stage Anchor-Free Approach for Part-Body Association", *International Conference on Learning Representations* (***ICLR***), 2024.
- R. Xia, C. Zhao, **M. Zheng**, Z. Wu, Q. Sun and Y. Tang, "CMDA: Cross-Modality Domain Adaptation for Nighttime Semantic Segmentation", *IEEE/CVF International Conference on Computer Vision* (***ICCV***), 2023.
- X. Gong, L. Song, **M. Zheng**, B. Planche, T. Chen, J. Yuan, D. Doermann and Z. Wu, "Progressive Multi-view Human Mesh Recovery with Self-Supervision", *Association for the Advancement of Artificial Intelligence* (***AAAI***), ***oral***, 2023.
- **M. Zheng**, B. Planche, X. Gong, F. Yang, T. Chen and Z. Wu, "Self-supervised 3D Patient Modeling with Multi-modal Attentive Fusion", *International Conference on Medical Image Computing and Computer Assisted Intervention* (***MICCAI***), 2022. **Early accept** (top 13% of submissions received a provisional accept recommendation).
- **M. Zheng**, S. Karanam, T. Chen, R. J. Radke and Z. Wu. "Visual Similarity Attention", *International Joint Conference on Artificial Intelligence* (***IJCAI***), 2022.
- X. Gong, **M. Zheng**, B. Planche, S. Karanam, T. Chen, D. Doermann and Z. Wu, "Self-supervised Human Mesh Recovery with Cross-Representation Alignment", *European Conference on Computer Vision* (***ECCV***), 2022.
- Q. Liu, **M. Zheng**, B. Planche, S. Karanam, T. Chen, M. Niethammer and Z. Wu, "PseudoClick: Interactive Image Segmentation with Click Imitation", *European Conference on Computer Vision* (***ECCV***), 2022.
- H. Guo, B. Planche, **M. Zheng**, S. Karanam, T. Chen, Z. Wu, "SMPL-A: Modeling Person-Specific Deformable Anatomy", *IEEE Conference on Computer Vision and Pattern Recognition* (***CVPR***), 2022.
- L. Song, X. Gong, B. Planche, **M. Zheng**, D. Doermann, J. Yuan, T. Chen, and Z. Wu, "PREF: Predictability Regularized Neural Motion Fields", *European Conference on Computer Vision* (***ECCV***), **oral**, 2022.
- F. Yang, S. Karanam, **M. Zheng**, T. Chen, H. Ling, Z. Wu, "Multi-Motion and Appearance Self-Supervised Moving Object Detection", *IEEE/CVF Winter Conference on Applications of Computer Vision* (***WACV***), 2022.
- R. Li, S. Karanam, **M. Zheng**, T. Chen and Z. Wu, "Everybody Is Unique: Towards Unbiased Human Mesh Recovery", *The British Machine Vision Conference* (***BMVC***), **oral**, 2021.
- A. Aich, **M. Zheng**, S. Karanam, T. Chen, A. K. Roy-Chowdhury and Z. Wu, "Spatio-temporal representation factorization for video-based person re-identification", *IEEE/CVF International Conference on Computer Vision* (***ICCV***), 2021.
- Y. Ge, Y. Xiao, Z. Xu, **M. Zheng**, S. Karanam, T. Chen, L. Itti and Z. Wu, "A Peek Into the Reasoning of Neural Networks: Interpreting with Structural Visual Concepts", *IEEE Conference on Computer Vision and Pattern Recognition* (***CVPR***), 2021.
- W. Liu, R. Li, **M. Zheng**, S. Karanam, Z. Wu, B. Bhanu, R. J. Radke and O. Camps. "Towards Visually Explaining Variational Autoencoders", *IEEE Conference on Computer Vision and Pattern Recognition* (***CVPR***), 2020.

- **M. Zheng**, S. Karanam, Z. Wu, R. J. Radke. "Re-Identification with Consistent Attentive Siamese Networks", *IEEE Conference on Computer Vision and Pattern Recognition* (***CVPR***), 2019.
- **M. Zheng**, S. Karanam, R. J. Radke. "RPIfield: A New Dataset for Temporally Evaluating Person Re-Identification", *IEEE Conference on Computer Vision and Pattern Recognition Workshops* (***CVPR-W***), 2018.
- **M. Zheng**, W. Liu, Y. Xiang. "Subdomain clutter reduction method for imaging near surface target", *IET International Radar Conference*, 2015.
- W. Liu, **M. Zheng**, Y. Xiang. "An autofocusing imaging algorithm for wall penetrating SAR", *IET International Radar Conference*, 2015.
- W. Liu, W. Wei, **M. Zheng**. "A near-field back-projection algorithm for wall penetrating SAR", *IET International Radar Conference*, 2015.
- X. Wang, H. Liu, **M. Zheng**. "A clutter suppression method of ground penetrating radar for detecting shallow surface target", *IET International Radar Conference*, 2015.

AWARDS

- Allen B. Dumont Prize – Rensselaer Polytechnic Institute, 2020. This prize is awarded to a graduate student who has demonstrated high scholastic ability and has made a substantial contribution to that field.
- Outstanding Graduates – Beijing Institute of Technology, 2013.
- Outstanding Student – Beijing Institute of Technology, 2012.
- Outstanding Student Scholarship – Beijing Institute of Technology, 2009, 2010, 2011, 2012.
- Talent Class Scholarship – Beijing Institute of Technology, 2009, 2010, 2011, 2012.
- Excellent Class Performance scholarship – Beijing Institute of Technology, 2009, 2010, 2011.

PATENTS

- **Active**
  - **M. Zheng**, S. Karanam, Z. Wu, A. Innanje, T. Chen, "Systems and methods for annotating 3D data", US12444171B2, 2025.
  - S. Karanam, **M. Zheng**, Z. Wu, "Automatic patient positioning for medical equipment", US12582484B2, 2026.
  - B. Planche, Z. Cai, Z. Wu, **M. Zheng**, T. Chen, "Systems and methods for anonymizing images", US12488147B2, 2025.
  - B. Planche, L. Song, Z. Wu, **M. Zheng**, "Systems and methods for motion estimation and view prediction", US12450755B2, 2025.
  - B. Planche, P. Sibut-Bourde, Z. Wu, **M. Zheng**, Z. GAO, A. Sharma, "Systems and methods for determining anatomical deformations", US12437401B2, 2025.
  - B. Planche, Z. Wu, **M. Zheng**, A. Sharma, "Systems and methods for determining 3D human pose based on 2D keypoints", US12394087B2, 2025.
  - A. Innanje, A. Sharma, B. Planche, **M. Zheng**, S. Sun, Z. Wu, T. Chen, "Patient care record management system", US12361161B2, 2025.
  - **M. Zheng**, E. Zhao, S. Karanam, Z. Wu, T. Chen, "Interactive Contour Refinements for Data Annotation", US12232900B2, 2025.
  - Z. Wu, S. Karanam, **M. Zheng**, A. Sharma, "Automated collision avoidance in medical environments", US12186913B2, 2025.
  - S. Karanam, **M. Zheng**, Z. Wu, "Systems and methods for human model recovery", US12183019B2, 2024.
  - **M. Zheng**, S. Karanam, Z. Wu, "Human model recovery using deep learning techniques", US12136235B2, 2024.

- B. Planche, Z. Wu, **M. Zheng**, "Systems and methods for recording medical environments", US12014815B2, 2024.
- S. Karanam, **M. Zheng**, Z. Wu, "Multi-view patient model construction", US11948250B2, 2024.
- S. Karanam, **M. Zheng**, Z. Wu, "Systems and methods for personalized patient body modeling", US11941738B2, 2024.
- **M. Zheng**, A. Sharma, S. Karanam, Z. Wu, "Automated patient modeling and positioning", US11896408B2, 2023.
- Z. Wu, S. Karanam, **M. Zheng**, A. Sharma, "Automatic organ geometry determination", US12051204B2, 2023.
- Z. Wu, S. Karanam, **M. Zheng**, A. Sharma, R. Li, "Automating a medical environment", US20220125400A1, 2022.
- A. Sharma, **M. Zheng**, S. Karanam, Z. Wu, A. Innanje, T. Chen, "Abnormality detection within a defined area", US20210272258A1, 2022.

Published (Pending)

- B. Planche, Z. Wu, **M. Zheng**, Z. Gao, A. Sharma, *et al.*, "Systems and methods for automatic medical report generation", US20250285718A1, 2025.
- Z. Wu, **M. Zheng**, B. Planche, Z. Gao, A. Sharma, T. Chen, "Systems and methods for explaining machine learning models", US20250259436A1, 2025.
- Z. Gao, A. Sharma, **M. Zheng**, B. Planche, Z. Wu, T. Chen, F. Yang, Y. Liu, "Systems and methods for automatic hand gesture recognition", US20250218222A1, 2025.
- Z. Gao, A. Sharma, **M. Zheng**, B. Planche, Z. Wu, Y. Liu, F. Yang, T. Chen, "Detection of body part and associated body in an image", US20250182517A1, 2025.
- B. Planche, Z. Wu, **M. Zheng**, Z. Gao, A. Sharma, "Predicting a position of an object over time", US20250117959A1, 2025.
- W. Cui, **M. Zheng**, A. Innanje, Z. Wu, T. Chen, "Segmentation of a region of interest in a medical image", US20250111517A1, 2025.
- **M. Zheng**, Z. Wu, B. Planche, Z. Gao, T. Chen, "Systems and methods for language-guided image retrieval", US20250094484A1, 2025.
- **M. Zheng**, X. Gong, B. Planche, Z. Wu, "Systems and methods for 3d human model estimation", US20240412452A1, 2024.
- Z. Gao, A. Sharma, **M. Zheng**, B. Planche, Z. Wu, T. Chen, "Motion detection associated with a body part", US20240378731A1, 2024.
- **M. Zheng**, J. Wang, B. Planche, Z. Gao, Z. Wu, "Systems and methods for multi-person pose estimation", US20240346684A1, 2024.
- Z. Wu, S. Karanam, **M. Zheng**, A. Sharma, "Automated calibration and validation of sensor network", US12026913B2, 2024.
- S. Karanam, R. Li, **M. Zheng**, Z. Wu, "Human model recovery based on video sequences", US20240177420A1, 2024.
- **M. Zheng**, W. Cui, Z. Wu, A. Innanje, B. Planche, T. Chen, "Systems and methods for automatic data annotation", US20240135737A1, 2024.
- **M. Zheng**, B. Planche, Z. Wu, T. Chen, "Systems and methods for surgical task automation", US20240099774A1, 2024.

ACADEMIC EXPERIENCES

- **Researcher**, U.S. Department of Homeland Security, Center of Excellence - SENTRY, June 2023 - present.
  - Expect to research and design anomaly detection and threat risk assessment algorithms for real-world homeland security applications.
- **Affiliated Student Researcher**, U.S. Department of Homeland Security, Center of Excellence - ALERT, August 2018 to May 2020.
  - Member of the Human Detection & Re-Identification for Mass Transit Environments project.
  - Research and design human re-identification algorithms for real-world homeland security applications.

SYNERGISTIC ACTIVITIES

- **Area Chair**
  - International Conference on Learning Representations (ICLR)
  - Advances in Neural Information Processing Systems (NeurIPS). Recognized as a **Top Area Chair** of NeurIPS 2026.
- **Reviewer**
  - IEEE Signal Processing Letters
  - IEEE Transactions on Neural Networks and Learning Systems
  - IEEE Transactions on Circuits and Systems for Video Technology
  - IEEE Access
  - International Journal of Computer Vision (IJCV)
  - European Conference on Computer Vision (ECCV). Recognized as an **Outstanding Reviewer** of ECCV 2026.
  - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
  - IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Recognized as an **Outstanding Reviewer** of CVPR 2026.
  - IEEE/CVF International Conference on Computer Vision (ICCV)
  - International Conference on Learning Representations (ICLR).
  - Advances in Neural Information Processing Systems (NeurIPS)
  - International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)
  - AAAI Conference on Artificial Intelligence (AAAI)
  - IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshop
- **Workshop Organizer**
  - Served as organizer of Workshop on Advanced Perception for Autonomous Healthcare, in conjunction with IEEE/CVF International Conference on Computer Vision (ICCV) 2025
- **Program Committee**
  - Artificial Intelligence with Biased or Scarce Data (AIBSD), in conjunction with 36th AAAI Conference on Artificial Intelligence 2022.